LAB 10
Art Gallery Management App

The **Art Gallery Management App** helps users track and manage their personal **art collection or gallery inventory**. The application allows users to perform **CRUD operations (Create, Read, Update, Delete)** on artwork entries.
All data must be stored locally using **SQLite**, and the app must run on both **Android and iOS via Expo**.

Data Structure
Each **artwork** includes the following attributes:

Field	Type	Description
id	integer	Unique identifier (auto-incremented by SQLite)
artwork_title	text	Title of the artwork (e.g., "Starry Night")
artist	text	Name of the artist (e.g., "Vincent van Gogh")
category	text	Category of the artwork (e.g., "Painting", "Sculpture", "Digital Art")
year_created	integer	Year the artwork was created
rating	real	User rating from **0.0 to 5.0**
favorite	integer	Favorite status (**0 = not favorite, 1 = favorite**)

General Requirements
• **Technology:** Use **Expo** to build the application
• **Data Storage:** Use **SQLite** for local storage
• **Navigation:** Use **Bottom Tab Navigator**
• **UI:** Clean and functional interface with proper styling

Bottom Tab Navigation Structure
The application must include **4 tabs**:
1. **Gallery** – Display all artworks
2. **Add Artwork** – Add new artwork
3. **Statistics** – Gallery statistics
4. **Favorites** – Favorite artworks and statistics

Task 1: Display Artwork List (Gallery Tab)
Create the **main screen** displaying artworks stored in SQLite.
Requirements:
• Use **FlatList** to display artworks
Each item should show:
- Artwork Title
- Artist
- Category
- Year Created
- Rating
- Favorite icon if marked favorite

• Display header:
"Art Gallery Manager"
• Implement **pull-to-refresh** to reload artworks.
• When tapping an artwork → navigate to **Artwork Detail Screen**.

Task 2: Add New Artwork (Add Artwork Tab)
Create a screen to **add new artwork entries**.

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The form must contain:

- Artwork Title
- Artist
- Category
- Year Created
- Rating
- Favorite (Switch / Checkbox)

Features:
• **Save button** inserts new artwork into SQLite
• Show **success message** after saving
• Automatically **navigate back to Gallery screen**

Task 3: Artwork Detail View with Actions
Create a **detail screen** showing full artwork information.
Displayed fields:

- Artwork Title
- Artist
- Category
- Year Created
- Rating
- Favorite status

Include two buttons:
Edit Button
• Navigates to edit screen with **pre-filled data**
• Button label changes **Edit → Save**
• Update artwork information in SQLite.
Delete Button
• Show confirmation dialog:
Example:
"Are you sure you want to delete this artwork?"
• If confirmed → delete artwork from database.

Task 4: Search and Filter Artworks
Implement **search and filtering functionality** in the **Gallery Tab**.
4.1 Search by Artwork Title
• Add a **Search Bar** at the top of the Gallery screen
• Implement **real-time search**
Example:
Typing **"star"** should return:
`Starry Night`

4.2 Search by Artist Name
Allow users to search artworks by **artist**.
Example:
Searching **"van Gogh"** should display:
`Starry Night`

4.3 Filter by Category
Provide a **category filter dropdown**.
Example categories:

- Painting
- Sculpture

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- Pop Art
- Digital Art

Users can filter artworks by category.
Example:
```
Category: Sculpture
```
Display only sculptures.

4.4 Filter Favorite Artworks
Add a **toggle filter**:
```
Show Favorites Only
```
When enabled:
• Only artworks with **favorite = 1** are displayed.

4.5 Reset Filters
Provide a **Reset button** to clear:
- Search keyword
- Category filter
- Favorite filter

The list should return to displaying **all artworks**.

Task 5: Gallery Statistics (Statistics Tab)
Create a **Statistics screen** displaying gallery insights.
Display the following:
Total number of artworks
Example:
```
Total Artworks: 10
```

Average rating
Example:
```
Average Rating: 4.67
```

Artwork count by category
Example:
```
Painting: 6
Sculpture: 3
Pop Art: 1
```

Task 6: Favorite Artworks (Favorites Tab)
Create a **Favorites screen**.
Display Favorite Artworks
Query artworks where:
```
favorite = 1
```
Display using **FlatList**.
Each item should display:
- Artwork Title
- Artist
- Category
- Rating

Favorite Statistics
Display statistics for favorite artworks.
Total favorites

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```
Total Favorites: 4
```
Average favorite rating
```
Average Favorite Rating: 4.85
```
Favorite artworks by category
Example:
```
Painting: 2
Sculpture: 1
Pop Art: 1
```

Sample Database Data

ID	Artwork Title	Artist	Category	Year Created	Rating	Favorite
1	Starry Night	Vincent van Gogh	Painting	1889	4.9	1
2	Mona Lisa	Leonardo da Vinci	Painting	1503	5.0	1
3	The Persistence of Memory	Salvador Dalí	Painting	1931	4.7	0
4	The Thinker	Auguste Rodin	Sculpture	1904	4.6	0
5	The Scream	Edvard Munch	Painting	1893	4.8	1
6	Girl with a Pearl Earring	Johannes Vermeer	Painting	1665	4.7	0
7	David	Michelangelo	Sculpture	1504	4.9	1
8	The Kiss	Gustav Klimt	Painting	1908	4.6	0
9	Campbell's Soup Cans	Andy Warhol	Pop Art	1962	4.4	0
10	Balloon Dog	Jeff Koons	Sculpture	1994	4.3	0

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